EE180 Fall Homework 1 — Report

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How to run the code

All paths in the code are relative to the package root. Figures are written to `image/` and the two prediction files are written into the package root.

```
python -m pip install -r requirements.txt
python codes/q1_analysis.py    # Q1 figures + measurements
python codes/q2_analysis.py    # Q2 figures + measurements
python codes/q3_classification.py  # Q3 figures + prediction CSVs
```

The three scripts are self-contained and share `codes/hw1_common.py`. `codes/HW1.ipynb` contains the same analysis interactively. Every time–frequency plot follows the required convention $M_{\mathrm{dB}}(t, f) = 20 \log_{10} |X|$ with a common magnitude reference and a positive floor of 10^{-6} before the logarithm. The shared reference is stated in each figure caption.

1 Question 1: Fourier and Time–Frequency Analysis

(a) Time domain and FFT

The two signals are 4800 samples long at $f_s = 60\,\mathrm{MHz}$ ($80\,\mu\mathrm{s}$ total). The FFT uses a single-sided amplitude normalization: $|X[k]|$ is divided by the record length N, positive-frequency interior bins are doubled, and the DC bin is not doubled. Both spectra are displayed in dB with the same reference (the larger of the two amplitude maxima) and the same axis limits (Fig. 1).

The time-domain signals look like band-limited noise bursts of almost identical average power; only the envelope timing differs (A: $24.7\,\mu\mathrm{s}$, B: $55.3\,\mu\mathrm{s}$). The FFT magnitude spectra are nearly identical: two sharp lines at $3.800\,\mathrm{MHz}$ and $4.200\,\mathrm{MHz}$, a broad raised region from ≈ 4.9 to $\approx 7.1\,\mathrm{MHz}$, a line at $\approx 9.0\,\mathrm{MHz}$, and an otherwise similar noise floor.

The FFT magnitude alone cannot distinguish the temporal structures because it is a time-averaged, phase-independent decomposition. Both recordings contain the *same* set of stationary spectral components, but arranged with different time evolution. The magnitude spectrum discards the phase, so the chirp direction, the chirp endpoints, and the arrival time of the 9 MHzcomponent are lost. In particular, the linear chirp spreads its energy over its whole sweep, and its contribution and the fixed tones overlap in frequency, which makes the spectral magnitudes of A and B look alike.

(b) STFT

STFT parameters are exactly as required: Hann window of 240 samples ($4\,\mu\mathrm{s}$), hop 6 samples ($0.1\,\mu\mathrm{s}$), $N_{\mathrm{FFT}} = 4096$, no detrending. Boundary handling: `boundary=None` and `padded=False`, i.e.

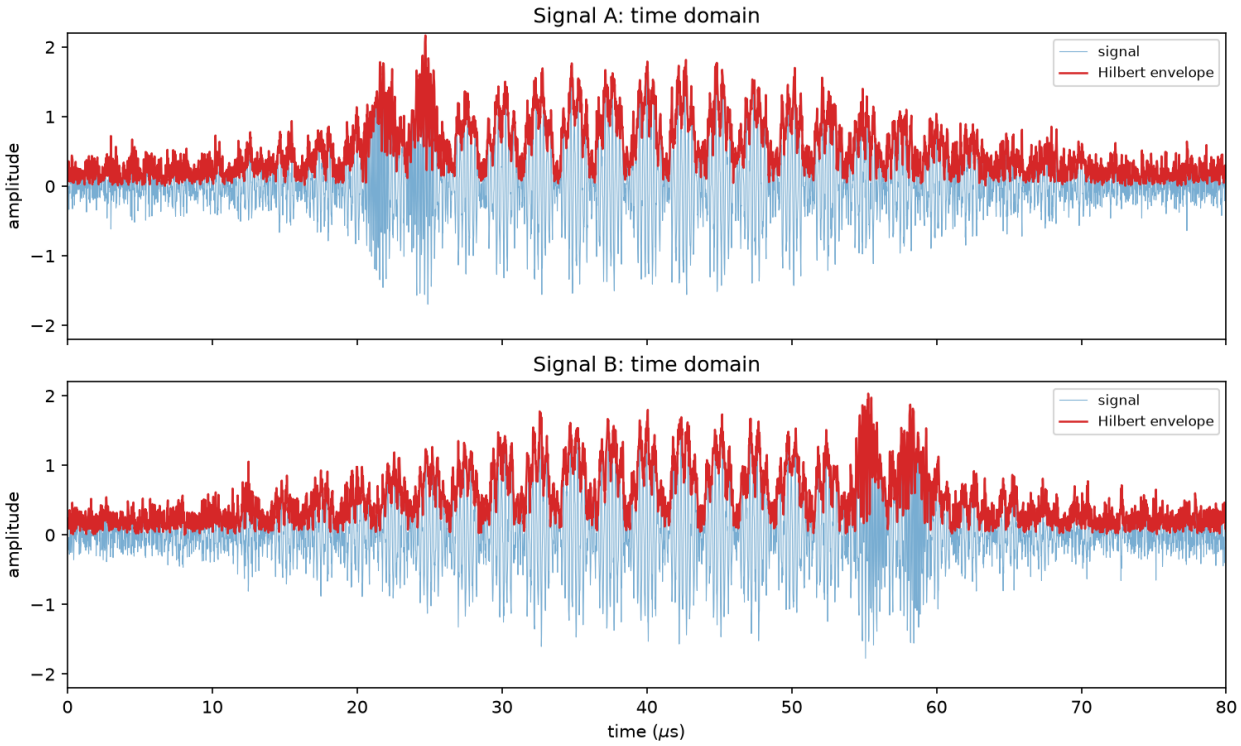


Figure 1: Q1(a): time-domain waveforms with the Hilbert envelope overlaid (same limits for both panels). Signal A peaks early (envelope maximum at 24.7 μs); signal B peaks late (55.3 μs). Both have nearly equal RMS amplitude (0.472 vs. 0.472).

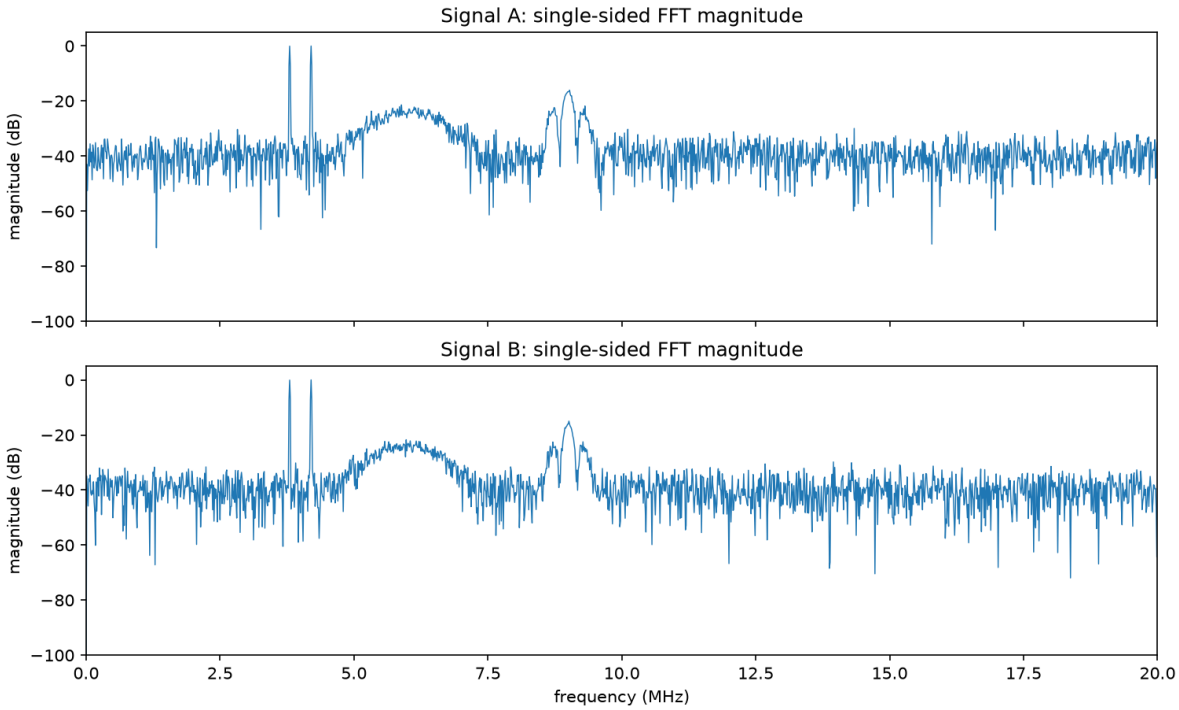


Figure 2: Q1(a): single-sided FFT magnitude, common normalization and common dB reference, identical axis limits. The two spectra are nearly indistinguishable.

no signal extension; only windows that fit entirely inside the record are analyzed, so the first and last 2 μs are not covered. The two spectrograms share one magnitude reference (the largest STFT magnitude of either signal) and one color scale (-80 to 0 dB) (Fig. 3).

The STFT reveals that the two signals are built from the same three components placed differently in time:

1. Two steady tones at 3.800 MHz and 4.200 MHz (separation 400 kHz), present over the whole record in both signals.
2. A linear chirp. In A it sweeps *up* from $\approx$ 4.9 MHz ($\approx$ 4 μs) to $\approx$ 7.1 MHz ($\approx$ 76 μs), slope $\approx +30\,\mathrm{GHz}/s$; in B it sweeps *down* from $\approx$ 7.1 MHz to $\approx$ 4.9 MHz with $\approx -30\,\mathrm{GHz}/s$.
3. A narrowband component at 9.02 MHz (A) / 9.01 MHz (B) that appears as a *pair* of short pulses: at 21.5 μs and 24.5 μs for A, and at 55.5 μs and 58.5 μs for B (lobe separation 2.97 μs, pulse widths $\approx$ 1.7 μs at -6 dB).

Thus the temporal order of the chirp and the timing of the 9 MHz pulse pair are the features that distinguish A from B, and they are directly visible in the STFT as the opposite slope of the diagonal ridge and the different locations of the 9 MHz blobs.

2 Question 2: STFT, CWT, and Time–Frequency Resolution

(a) Short and long windows and the signal components

Two representative windows were used, both with a Hann window, hop 8 samples (0.133 μs), $N_{\mathrm{FFT}} = 4096$, and no boundary extension:

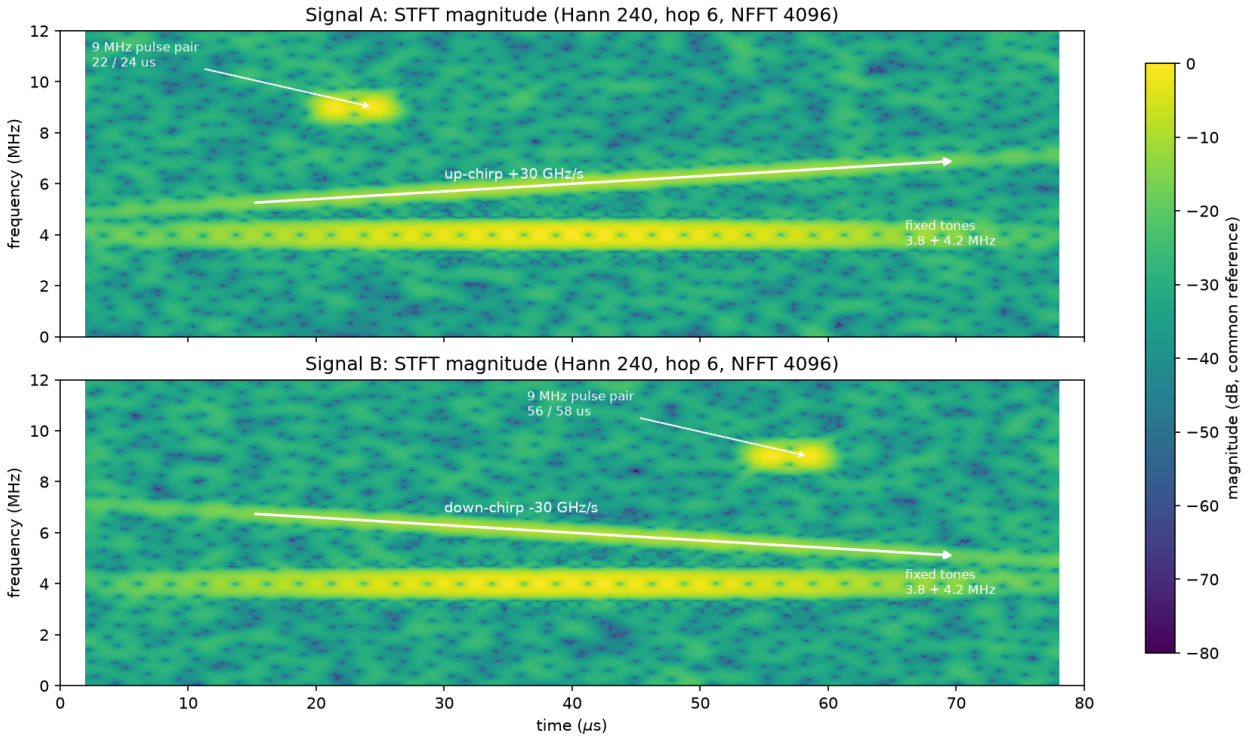


Figure 3: Q1(b): required-parameter STFT of both signals, common normalization and color scale. The chirp direction and the pulse-pair time are the discriminating features.

- **Short:** 128 samples (2.13 μs). Hann main-lobe width $4f_s/N \approx 1.88$ MHz. Gives good time localization.
- **Long:** 1024 samples (17.07 μs). Hann main-lobe width ≈ 0.234 MHz. Gives good frequency localization.

Both spectrograms in Fig. 4 use the same reference (largest magnitude across the two) and the same $-80 \ldots 0$ dB color scale.

Components of Signal A.

1. **Two steady tones** at 3.800 MHz and 4.200 MHz, separation 0.400 MHz, present for the full duration. The separation is obtained from the long-window spectrogram (and confirmed by the full-record FFT); the short-window main lobe (1.88 MHz) merges them into a single 4 MHz band.
2. **Linear up-chirp** from ≈ 4.9 MHz at $\approx 4\,\mu$s to ≈ 7.1 MHz at $\approx 76\,\mu$s. A line fit to the ridge gives $f(t) = 4.84 + 0.029\,t$ (MHz, μs), i.e. $+(29\text{–}30)$ GHz/s; the slight spread of the estimate with window length indicates a small non-linearity. The distance between the chirp and the upper tone grows with time: 0.78 MHz at 5 μs, 1.22 MHz at 20 μs, 1.95 MHz at 45 μs, and 2.82 MHz at 75 μs. The long window resolves the chirp from the tones at all these times; the short window cannot separate them below $\approx 20\,\mu$s (Fig. 5).
3. **9 MHz pulse pair** centered at 9.02 MHz with lobes at 21.5 μs and 24.5 μs (separation 2.97 μs, each lobe $\approx 1.7\,\mu$s long at -6 dB). The short window resolves the two lobes; the long window smears them into one blob (Fig. 4, lower panel).

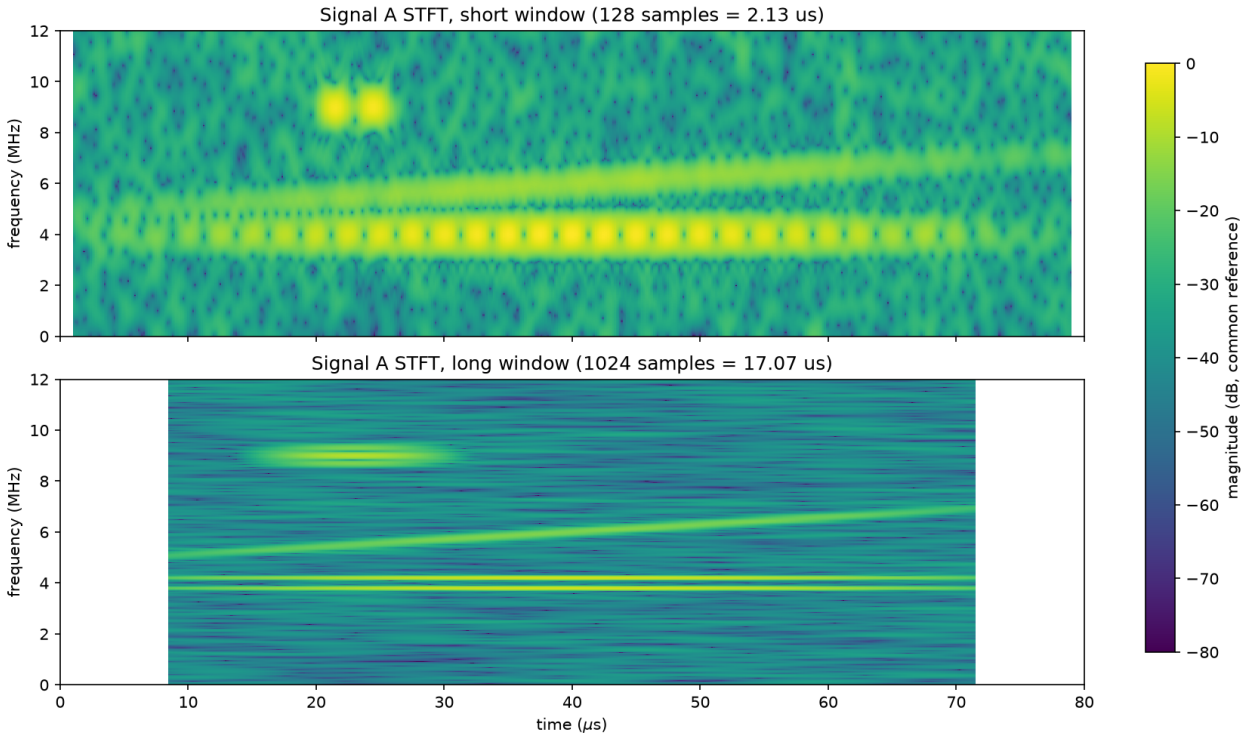


Figure 4: Q2(a): Signal A STFT with short and long windows, common normalization and color scale. The chirp is visible in both; the low-frequency tone doublet is only resolved by the long window.

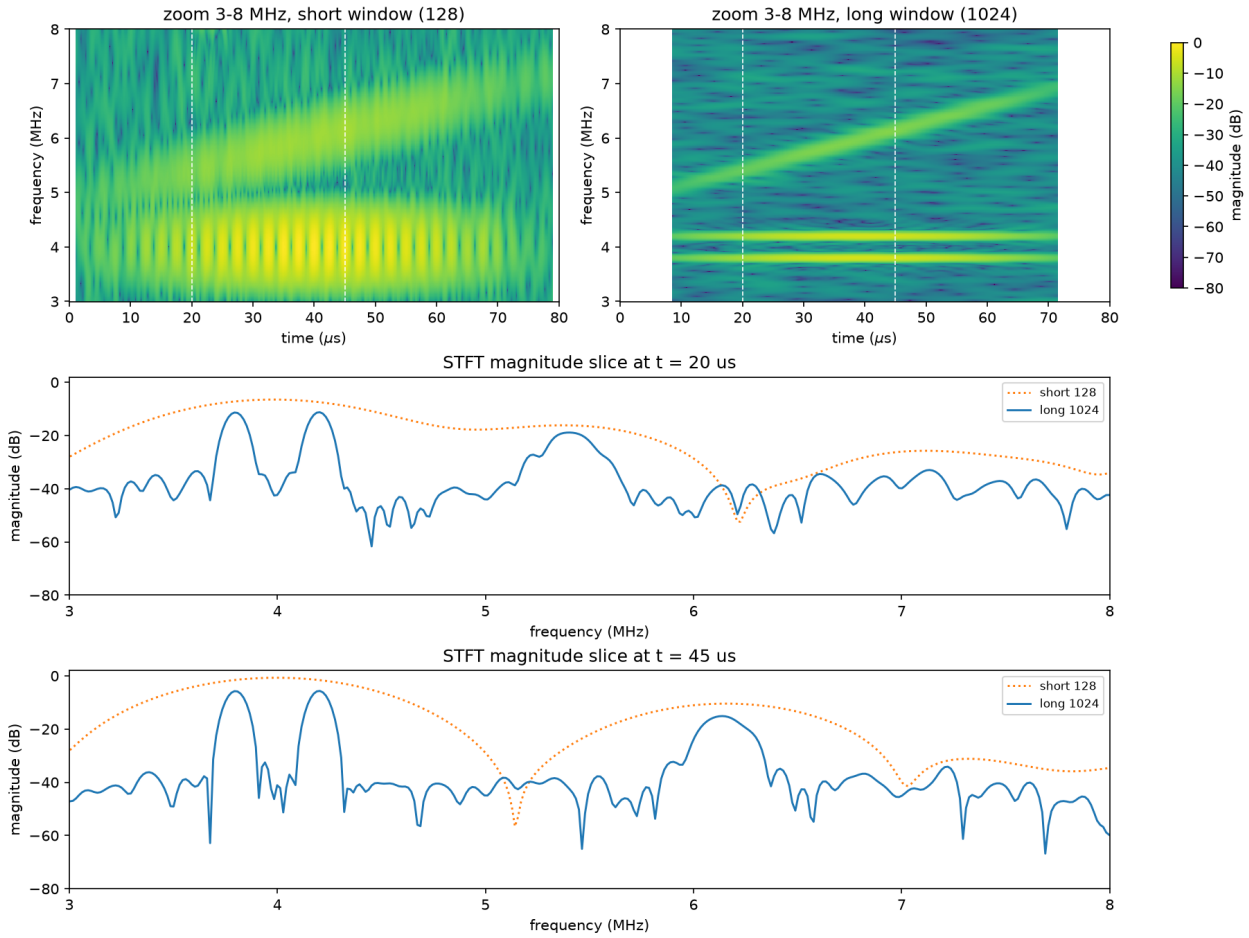


Figure 5: Q2(a): 3–8 MHz zooms (top) and 1-D STFT slices at 20 and 45 μs (bottom). The long window separates 3.800/4.200 MHz and the chirp ridge; the short window merges them into one broad peak.

The slices in Fig. 5 support these estimates directly: at $t = 20\,\mu$s the long window shows three peaks (3.80, 4.20, and the chirp at 5.42 MHz), while the short window shows one broad peak; at $t = 45\,\mu$s the long window shows the tones and the chirp at 6.15 MHz.

(b) Time–frequency resolution trade-off

For a Hann window the time and frequency widths satisfy $\Delta t\,\Delta f \approx 4$ (samples·bins): Δt scales as N/f_s while the main-lobe width scales as $4f_s/N$, so the product is fixed. For the two windows used here:

$$(2.13\,\mu\text{s})(1.88\,\text{MHz}) \approx 4.0, \qquad (17.07\,\mu\text{s})(0.234\,\text{MHz}) \approx 4.0.$$

The short window (4, top) localizes each 1.7 μs pulse and the chirp endpoints in time, but merges the 0.400 MHz tone doublet and the chirp with the tones at early times. The long window (bottom) separates the tones and the chirp, but merges the two 9 MHz lobes and smears the chirp onset/end by $\pm 8.5\,\mu$s.

No single window resolves all components: resolving the 0.400 MHz doublet requires a window of at least $\approx 4f_s/0.4\,\text{MHz} = 600$ samples (10 μs), which already merges the 3 μs-separated pulses; resolving the pulse pair cleanly requires $\Delta t \lesssim 1.5\,\mu$s ($N \lesssim 90$), which merges the tones and the early chirp. The two estimates therefore need the two spectrograms: frequencies and separations from the long window, times from the short window.

(c) Continuous wavelet transform

A complex Morlet wavelet `cmor1.5-1.0` (bandwidth $B = 1.5$, center frequency $C = 1.0$ cycles/sample at scale 1) was used with 400 log-spaced scales covering 2.5–12 MHz via $f = Cf_s/a$. PyWavelets returns L_2-normalized coefficients (unit-energy wavelet at each scale), whose magnitude for a constant-amplitude tone grows as $\sqrt{a}$; each scale row was therefore divided by $\sqrt{a}$ (per-scale L_1 normalization) so that equal-amplitude components have comparable brightness. The dB display uses the maximum compensated magnitude as reference, a 10^{-6} floor, and the $-80 \ldots 0$ dB scale (Fig. 6).

Comparison with the STFT. The complex Morlet has a constant quality factor $Q = f/\Delta f \approx 2.7$. Its Gaussian bandwidth is therefore ≈ 1.5 MHz at 4 MHz and ≈ 3.3 MHz at 9 MHz, while its effective time width is $\sigma_t \approx 0.87/f$: 0.22 μs at 4 MHz and 0.10 μs at 9 MHz (the product $\sigma_t\sigma_f$ is constant). The CWT result shows:

- the 9 MHz pulse pair is *resolved* in time (two lobes), like the short STFT and unlike the long STFT;
- the chirp is visible once it separates from the tones ($f_{\text{chirp}} - f_{\text{tone}} \gtrsim Q$-limited bandwidth), i.e. after $\approx$ 30–40 μs; at earlier times it merges with the tone band;
- the 3.800/4.200 MHz doublet is *not* resolved, because its 0.4 MHz separation is far below the 1.5 MHz bandwidth at 4 MHz; this is worse than the long STFT (0.234 MHz) and similar to the short STFT.

Does the CWT eliminate the trade-off? No. The CWT does not beat the uncertainty relation; at each frequency it merely chooses a different point on the Δt–Δf hyperbola: long effective windows at low frequency (good frequency resolution, coarse time) and short effective windows at high frequency (good time resolution, coarse frequency). This is why it resolves the high-frequency pulse pair but merges the low-frequency tone doublet at the same time. Resolving the 0.400 MHz

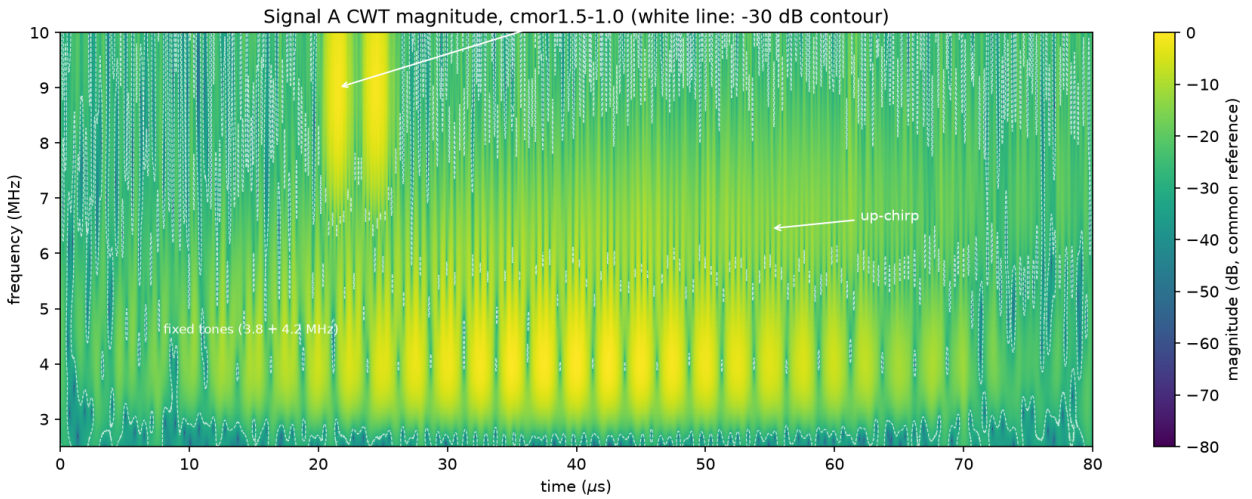


Figure 6: Q2(c): CWT magnitude of Signal A, `cmor1.5-1.0` (white line: −30 dB contour). The wavelet adapts its resolution with frequency: it resolves the two 9 MHz pulses, tracks the chirp once it is separated from the tones, but merges the 400 kHz tone doublet.

doublet at 4 MHz would require a much higher-Q analysis (e.g. a Morlet with large B, or the long STFT), which in turn would smear the 3 μs-separated pulses. The evidence across Figs. 4, 5 and 6 shows that the limitation is fundamental, not a property of a particular transform.

3 Question 3: Classifying Ultrasound Recordings

Each recording has 896 samples at $f_s = 26.5216$ MHz (33.8 μs). Features were computed identically for the reference, Stage 1, and Stage 2 sets; no per-recording normalization was applied.

Features and reference-set statistics

The main feature is the amplitude envelope of the analytic signal (Hilbert transform), smoothed by a centered five-point average. From it we extract (all times in μs):

- a_{main}: global envelope maximum;
- a_{early}, t_{early}: maximum envelope in $[5, 22)$ μs and its time;
- a_{late}: maximum envelope for $t \geq 22$ μs;
- late_ratio $= a_{\text{late}}/a_{\text{early}}$;
- doublet_ratio: maximum envelope in $[t_{\text{early}} + 2.1,\ t_{\text{early}} + 3.4]$ μs divided by a_{early}, which detects a second surface echo ≈ 2.75 μs after the main one.

The classes have clear physical signatures (Figs. 7 and 8):

- **Class 1 (acrylic):** a strong echo at ≈ 14.6 μs followed by a strong secondary echo 2.75 μs later (front and back surface of the plate), hence doublet_ratio 0.20–0.39.

Table 1: Reference-set feature statistics (mean ± std, $n = 15$ per class).

Feature	Class 1 (acrylic)	Class 2 (sandpaper)	Class 3 (kitchen paper)
a_{main}	1.072 ± 0.112	1.156 ± 0.083	0.519 ± 0.064
a_{early}	1.072 ± 0.112	1.156 ± 0.083	0.282 ± 0.118
a_{late}	0.061 ± 0.011	0.081 ± 0.030	0.478 ± 0.138
late_ratio	0.057 ± 0.011	0.069 ± 0.023	2.202 ± 1.375
t_{early} (μs)	14.63 ± 0.30	12.40 ± 0.12	11.85 ± 0.24
doublet_ratio	0.319 ± 0.041	0.015 ± 0.005	0.014 ± 0.009

- **Class 2 (sandpaper):** one dominant echo at $\approx 12.4\,\mu$s with no nearby secondary echo (doublet_ratio ≤ 0.023) and only weak late scattering ($a_{\text{late}} \leq 0.15$).
- **Class 3 (kitchen paper):** weak echoes overall ($a_{\text{main}} < 0.63$): an early echo at $\approx 12\,\mu$s and, in most recordings, a second weak echo at $\approx 26.5\,\mu$s (late_ratio up to ≈ 4.5). Within-class variation is larger here; two reference recordings have almost no late echo.

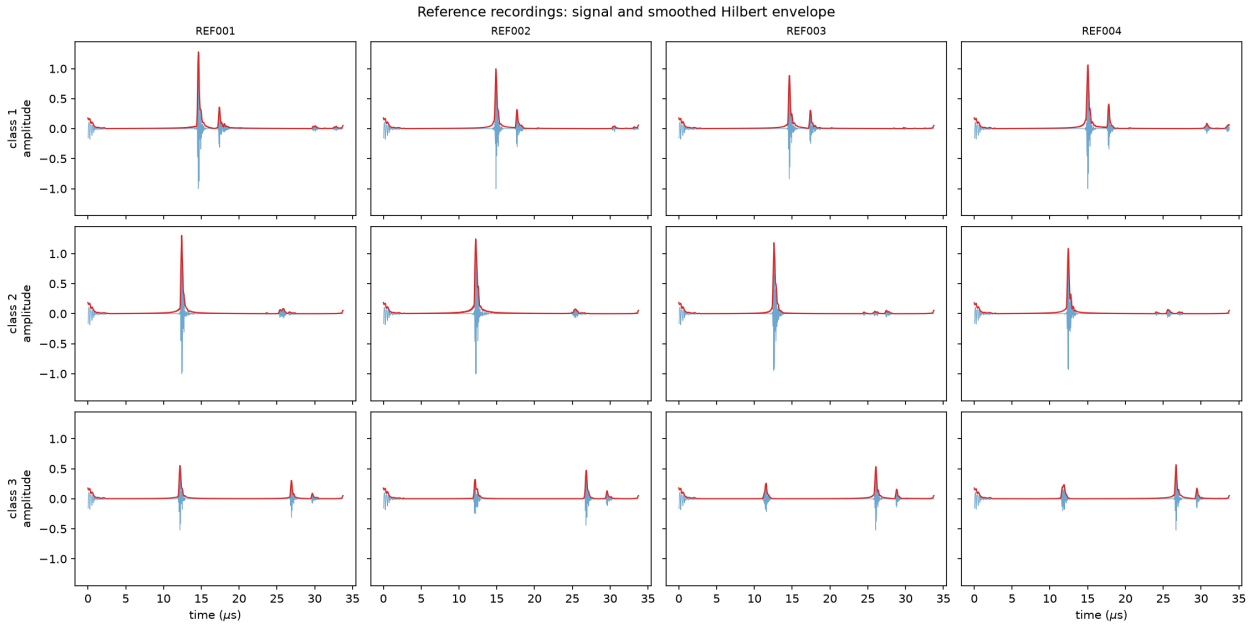


Figure 7: Q3: four reference recordings per class (blue: signal, red: smoothed Hilbert envelope). Class 1 shows a double echo; class 2 a single strong echo; class 3 weak early/late echoes.

Decision rules

The thresholds below were chosen from the labeled reference set only, using the visible gaps in Fig. 8:

1. **If doublet_ratio > 0.15: class 1.** Reference margin: class 1 minimum 0.201 vs. maximum 0.033 for classes 2 and 3.

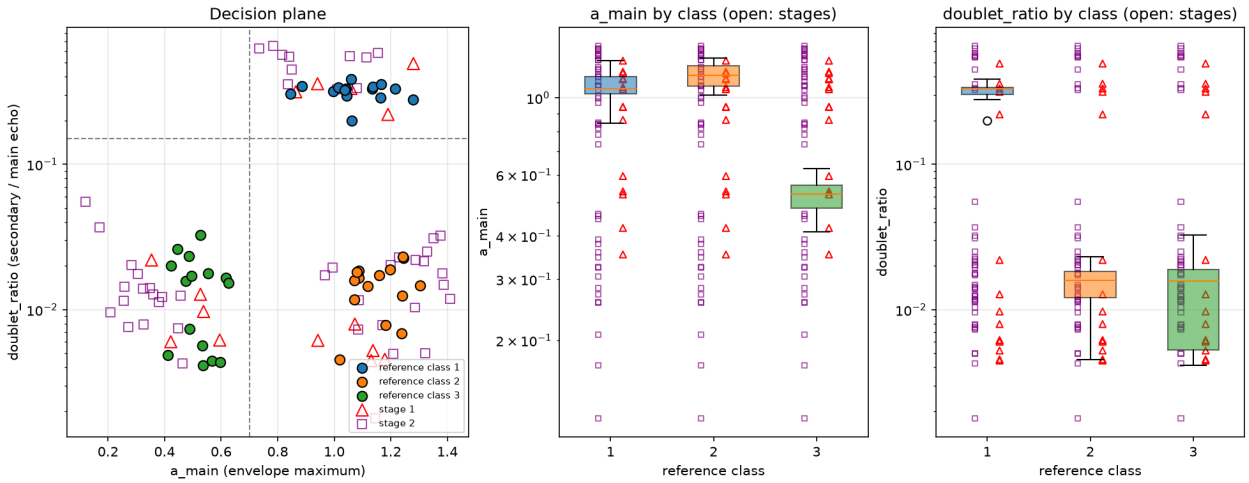


Figure 8: Q3: decision plane (left) with the two thresholds, and class-wise distributions of a_{main} and doublet_ratio including the Stage 1 (triangles) and Stage 2 (squares) recordings.

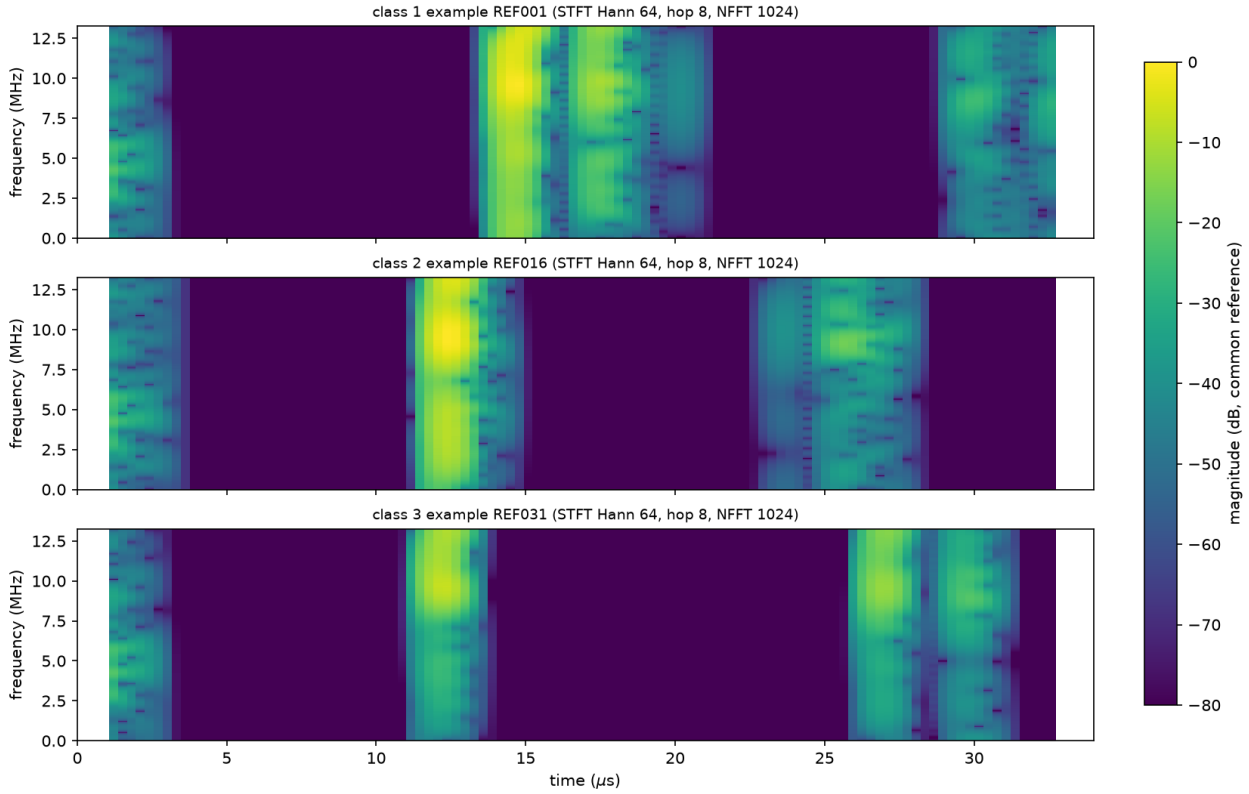


Figure 9: Q3: example STFT magnitude (Hann 64, hop 8, NFFT 1024) of one reference recording per class, common normalization and color scale (dB convention as required).

2. **Else if $a_{\mathbf{main}} < 0.70$: class 3.** Reference margin: class 3 maximum 0.626 vs. minimum 0.845 for classes 1 and 2.

3. **Otherwise: class 2.**

These explicit rules classify all 45 labeled reference recordings correctly (45/45, 100%). As an optional cross-check, a linear discriminant analysis on $[a_{\text{main}}, \text{late_ratio}, \text{doublet_ratio}]$ with leave-one-out evaluation also reaches 100%. The rules use ratios and echo timing, which are more robust to the handling differences of the classroom recordings than absolute amplitudes: in Stage 2 the strongest-echo time ranges from 10.3 to 16.1 μs, while the doublet pattern stays stable.

Task-stage predictions

Applying the same frozen rules to the anonymized sets (Eq. above; no labels used):

- **Stage 1** (`predictions_stage1.csv`, 15 recordings): 5 class 1, 5 class 2, 5 class 3. All 15 fall far from the thresholds, and the 5-per-class counts expected for this TA-collected set are reproduced.
- **Stage 2** (`predictions.csv`, 47 recordings): 11 class 1, 19 class 2, 17 class 3.

The class-3 group in Stage 2 includes both weak-echo patterns seen in the reference set (strong-late-echo and weak-early-echo-only), which is the main within-class variation observed. We do not report accuracy against the withheld labels, as required.